

the above equations we get another differential equation. Using this any one of the above equations as simultaneous equations in x, y , those can be solved using Cramer's rule. This gives two separate equations in x, y , which can be solved. Finally, using these values of x, y in the equations, we get relations among the arbitrary constants appeared in the solutions. Using these we express the arbitrary constants appeared in any of the variables with those appeared in other variable. Thus we get the values of x, y in which the number of arbitrary constants is equal to the order of differential equations.

WORKED-OUT EXAMPLES

1. Consider the system of differential equations:

$$\frac{dx}{dt} - 7x + y = 0; \quad \frac{dy}{dt} - 2x - 5y = 0.$$

Solution

Method 1. Differentiating the first equation, w.r.t. t , we get

$$\frac{d^2x}{dt^2} - 7\frac{dx}{dt} + \frac{dy}{dt} = 0,$$

Using the second equation, we get

$$\frac{d^2x}{dt^2} - 7\frac{dx}{dt} + 2x + 5y = 0.$$

Again using the first equation, we get

$$\frac{d^2x}{dt^2} - 12\frac{dx}{dt} + 37x = 0.$$

Putting $x = e^{mt}$, we get the auxiliary equation $m^2 - 12m + 37 = 0$, which gives $m = 6 \pm i$.

So
$$x = e^{6t} [A \cos t + B \sin t].$$

Differentiating this with respect to t , we get

$$\frac{dx}{dt} = 6x + e^{6t} [B \cos t - A \sin t].$$

Putting these values of $x, \frac{dx}{dt}$ in the first equation, we get

$$y = e^{6t} [(A - B) \cos t + (A + B) \sin t].$$

Hence, $x = e^{6t} [A \cos t + B \sin t]$ and $y = e^{6t} [(A - B) \cos t + (A + B) \sin t]$ are the required solution.

Method 2. We rewrite the system of equations as

$$(D - 7)x + y = 0; \quad -2x + (D - 5)y = 0, \quad \text{where } D = \frac{d}{dt}.$$

Eliminating y , we get

$$[(D-7)(D-5)+2]x=0 \Rightarrow [D^2-12D+37]x=0.$$

Auxiliary equation of the above equation is $m^2-12m+37=0$, which gives $m=6 \pm i$.

So $x = e^{6t}[A \cos t + B \sin t]$.

Putting this in the first equation, we get

$$y = e^{6t}[(A-B) \cos t + (A+B) \sin t].$$

Method 3. Rewriting the system of equation as

$$(D-7)x + y = 0; -2x + (D-5)y = 0, \text{ where } D = \frac{d}{dt}$$

and using Crammer's rule we get

$$\begin{vmatrix} D-7 & 1 \\ -2 & D-5 \end{vmatrix} x = \begin{vmatrix} 0 & 1 \\ 0 & D-5 \end{vmatrix} \text{ and } \begin{vmatrix} D-7 & 1 \\ -2 & D-5 \end{vmatrix} y = \begin{vmatrix} D-7 & 0 \\ -2 & 0 \end{vmatrix},$$

i.e., $[D^2-12D+37]x=0$ and $[D^2-12D+37]y=0$, which give

$$x = e^{6t}[A \cos t + B \sin t] \text{ and } y = e^{6t}[C \cos t + D \sin t].$$

Putting these values in $\frac{dx}{dt} - 7x + y = 0$, we get

$$[B-A+C] \cos t - [A+B-D] \sin t = 0,$$

which is true for all values of t . This implies

$$B-A+C=0 \text{ and } A+B-D=0.$$

Hence the solutions are

$$x = e^{6t}[A \cos t + B \sin t] \text{ and } y = e^{6t}[(A-B) \cos t + (A+B) \sin t].$$

2. Solve $\frac{dx}{dt} = 4x - 2y$, $\frac{dy}{dt} = x + y$.

Solution

Differentiating the first equation with respect to t , we get

$$\begin{aligned} \frac{d^2x}{dt^2} &= 4 \frac{dx}{dt} - 2 \frac{dy}{dt} = 4 \frac{dx}{dt} - 2(x+y), \text{ using the second equation} \\ &= 4 \frac{dx}{dt} - 2x - 4x + \frac{dx}{dt}, \text{ using the first equation} \end{aligned}$$

or, $\frac{d^2x}{dt^2} - 5 \frac{dx}{dt} + 6x = 0.$

Its auxiliary equation is $m^2 - 5m + 6 = 0$, whose solutions are $m = 2, 3$. So general solution of the above equation in x is

$$x = Ae^{2t} + Be^{3t}, \text{ where } A, B \text{ are arbitrary constants.}$$

Using this value of x in second equation we get

$$\frac{dy}{dt} - y = Ae^{2t} + Be^{3t}$$

which is a linear differential equation whose integrating factor is e^{-t} . Multiplying the above equation by e^{-t} , we get

$$\frac{d}{dt}(ye^{-t}) = Ae^t + Be^{2t}.$$

Integrating, we get

$$ye^{-t} = Ae^t + \frac{1}{2}Be^{2t} \text{ or, } y = Ae^{2t} + \frac{1}{2}Be^{3t}.$$

Hence $x = Ae^{2t} + Be^{3t}$ and $y = Ae^{2t} + \frac{1}{2}Be^{3t}$ are the required solutions.

3. Solve $\frac{dx}{dt} = y + e^t$; $\frac{dy}{dt} = -x + 1$.

Solution

From the first equation, we get

$$y = \frac{dx}{dt} - e^t.$$

Differentiating first equation with respect to t , we get

$$\frac{d^2x}{dt^2} = \frac{dy}{dt} + e^t = -x + 1 + e^t \text{ or, } (D^2 + 1)x = 1 + e^t, D = \frac{d}{dt}.$$

The roots of the auxiliary equation of the corresponding homogeneous equation are $\pm i$. So the complementary function is

$$x = A \cos t + B \sin t$$

and P.I. = $\frac{1}{D^2 + 1}(1 + e^t) = 1 + \frac{1}{2}e^t$. So the general solution of the above differential equation in x is

$$x = A \cos t + B \sin t + 1 + \frac{1}{2}e^t.$$

Using this in $y = \frac{dx}{dt} - e^t$, we get

$$y = -A \sin t + B \cos t - \frac{1}{2}e^t.$$

Hence, $x = A \cos t + B \sin t + 1 + \frac{1}{2}e^t$ and $y = -A \sin t + B \cos t - \frac{1}{2}e^t$ are the required solutions.

4. Solve the following system of equations: $\frac{dx}{dt} + 4x + 3y = t$; $\frac{dy}{dt} + 2x + 5y = e^t$.

Solution

Rewriting the system of equations as

$$(D + 4)x + 3y = t; 2x + (D + 5)y = e^t, \text{ where } D = \frac{d}{dt}.$$

Eliminating y , we get

$$(D+5)(D+4)x - 6x = (D+5)t - 3e^t,$$

$$\text{i.e., } [D^2 + 9D + 14]x = 1 + 5t - 3e^t.$$

Solution of this equation is

$$x = Ae^{-7t} + Be^{-2t} + \frac{1}{D^2 + 9D + 14}(1 + 5t - 3e^t)$$

$$= Ae^{-7t} + Be^{-2t} + \frac{1}{14}\left(1 + 5t - \frac{45}{14}\right) - \frac{1}{8}e^t$$

$$= Ae^{-7t} + Be^{-2t} + \frac{1}{14}\left(5t - \frac{31}{14}\right) - \frac{1}{8}e^t.$$

Differentiating this with respect to t , we get

$$\frac{dx}{dt} = -7Ae^{-7t} - 2Be^{-2t} + \frac{5}{14} - \frac{1}{8}e^t.$$

Using the values of $x, \frac{dx}{dt}$ in the first equation, we get

$$y = \frac{1}{3}\left(3Ae^{-7t} - 2Be^{-2t} - \frac{3t}{7} + \frac{5}{8}e^t + \frac{27}{98}\right).$$

Hence,

$$x = Ae^{-7t} + Be^{-2t} + \frac{1}{14}\left(5t - \frac{31}{14}\right) - \frac{1}{8}e^t$$

and

$$y = \frac{1}{3}\left(3Ae^{-7t} - 2Be^{-2t} - \frac{3t}{7} + \frac{5}{8}e^t + \frac{27}{98}\right)$$

are the required solutions.

5. Solve the following system of differential equation:

$$\frac{dx}{dt} + 5x + y = e^t; \quad \frac{dy}{dt} - x + 3y = e^{2t}.$$

Solution

Rewriting the system of equation as

$$(D+5)x + y = e^t; \quad -x + (D+3)y = e^{2t}, \text{ where } D = \frac{d}{dt}.$$

Eliminating y , we get

$$[(D+5)(D+3)+1]x = 4e^t - e^{2t},$$

$$\text{i.e., } [D^2 + 8D + 16]x = 4e^t - e^{2t}, \text{ which gives}$$

$$x = (A + Bt)e^{-4t} + \frac{1}{(D+4)^2}(4e^t - e^{2t}) = (A + Bt)e^{-4t} + \frac{4}{25}e^t - \frac{1}{36}e^{2t}.$$

Differentiating it with respect to t , we get

$$\frac{dx}{dt} = (B - 4A - 4Bt)e^{-4t} + \frac{4}{25}e^t - \frac{1}{18}e^{2t}.$$

Using the values of $x, \frac{dx}{dt}$ in the first equation we get

$$\begin{aligned} y &= e^t - \frac{dx}{dt} - 5x \\ &= e^t - (B - 4A - 4Bt)e^{-4t} - \frac{4}{25}e^t + \frac{1}{18}e^{2t} - 5\left[(A + Bt)e^{-4t} + \frac{4}{25}e^t - \frac{1}{36}e^{2t}\right] \\ &= -(A + B + Bt)e^{-4t} + \frac{1}{25}e^t + \frac{7}{36}e^{2t}. \end{aligned}$$

Hence, $x = (A + Bt)e^{-4t} + \frac{4}{25}e^t - \frac{1}{36}e^{2t}$

and $y = -(A + B + Bt)e^{-4t} + \frac{1}{25}e^t + \frac{7}{36}e^{2t}$ are the required solutions.

6. Solve the following system of differential equations:

$$2\frac{dx}{dt} + \frac{dy}{dt} - 4x - y = e^t; \quad \frac{dx}{dt} + 3x + y = 0.$$

Solution

Rewriting the system of equations as

$$2(D-2)x + (D-1)y = e^t; \quad (D+3)x + y = 0, \text{ where } D = \frac{d}{dt}.$$

Eliminating y we get

$$[2(D-2) - (D-1)(D+3)]x = e^t, \text{ i.e., } [D^2 + 1]x = -e^t,$$

which gives

$$x = A \cos t + B \sin t - \frac{1}{D^2 + 1}e^t = A \cos t + B \sin t - \frac{1}{2}e^t.$$

Putting this value of x in the second equation, we get

$$y = -(D+3)x = (A-3B)\sin t - (B+3A)\cos t + 2e^t.$$

Hence,

$$x = A \cos t + B \sin t - \frac{1}{2}e^t \text{ and } y = (A-3B)\sin t - (B+3A)\cos t + 2e^t$$

are the required solutions.

7. Solve the following system of differential equations:

$$\frac{dx}{dt} + \frac{dy}{dt} - 2y = 2 \cos t - 7 \sin t; \quad \frac{dx}{dt} - \frac{dy}{dt} + 2x = 4 \cos t - 3 \sin t.$$

Solution

We rewrite the equation as

$$Dx + (D-2)y = 2 \cos t - 7 \sin t; \quad (D+2)x - Dy = 4 \cos t - 3 \sin t.$$

Eliminating y , we get

$$[-D^2 - (D^2 - 4)]x = -D(2 \cos t - 7 \sin t) - (D-2)(4 \cos t - 3 \sin t)$$

$$(D^2 - 2)x = -9 \cos t, \text{ which gives } x = Ae^{\sqrt{2}t} + Be^{-\sqrt{2}t} + 3 \cos t.$$

Adding the given equations, we get

$$2(D+1)x - 2y = 6\cos t - 10\sin t$$

or,

$$y = (D+1)[Ae^{\sqrt{2}t} + Be^{-\sqrt{2}t} + 3\cos t] - 3\cos t + 5\sin t$$

$$= (1+\sqrt{2})Ae^{\sqrt{2}t} + (1-\sqrt{2})Be^{-\sqrt{2}t} + 2\sin t.$$

Hence, $x = Ae^{\sqrt{2}t} + Be^{-\sqrt{2}t} + 3\cos t$

and $y = (1+\sqrt{2})Ae^{\sqrt{2}t} + (1-\sqrt{2})Be^{-\sqrt{2}t} + 2\sin t$

are the required solutions.

8. Solve the following system of differential equations:

$$\frac{dx}{dt} + \frac{dy}{dt} + 2x + y = 0; \quad \frac{dy}{dt} + 5x + 3y = 0.$$

Solution

The given equations can be written as

$$(D+2)x + (D+1)y = 0,$$

$$5x + (D+3)y = 0.$$

Eliminating y , we get $(D^2+1)x = 0$.

Auxiliary equation of the above differential equation in x is $m^2 + 1 = 0$, which gives $m = \pm i$. So, the general solution of the above equation is

$$x = A\cos t + B\sin t, \text{ where } A, B \text{ are arbitrary constants.}$$

Differentiating it, we get

$$\frac{dx}{dt} = B\cos t - A\sin t. \text{ Subtracting the given equation, we get}$$

$$\frac{dx}{dt} = 3x + 2y$$

or, $2y = B\cos t - A\sin t - 3(A\cos t + B\sin t)$

or, $y = \frac{B-3A}{2}\cos t - \frac{A+3B}{2}\sin t.$

Hence $x = A\cos t + B\sin t$ and $y = \frac{B-3A}{2}\cos t - \frac{A+3B}{2}\sin t$ are the required solutions.

9. Solve the following system of differential equations:

$$\frac{d^2x}{dt^2} - 2x - 3y = e^{2t}; \quad \frac{d^2y}{dt^2} + x + 2y = 0.$$

Solution

The given equations can be written as

$$[D^2 - 2]x - 3y = e^{2t}; \quad x + [D^2 + 2]y = 0.$$

Solving the

which gives x

Putting this

Hence,

are the required

10. Solve the

Solution

The given

Eliminat

Auxiliar

which gives

Hence

Putting

Hence,

and

11. Solve

Solution

From

Solving the above system for x , we get

$$[(D^2 - 2)(D^2 + 2) + 3]x = (D^2 + 2)e^{2t} \quad [D^4 - 1]x = 6e^{2t},$$

which gives $x = A \cos t + B \sin t + Ee^t + Fe^{-t} + \frac{2}{5}e^{2t}$.

Putting this value of x in the first equation, we get

$$y = -(A \cos t + B \sin t) - \frac{1}{3}(Ee^t + Fe^{-t}) - \frac{1}{15}e^{2t}.$$

Hence, $x = A \cos t + B \sin t + Ee^t + Fe^{-t} + \frac{2}{5}e^{2t}$ and

$$y = -(A \cos t + B \sin t) - \frac{1}{3}(Ee^t + Fe^{-t}) - \frac{1}{15}e^{2t}$$

are the required solutions.

10. Solve the following system of differential equations:

$$\frac{d^2x}{dt^2} - 3x - 4y = 0; \quad \frac{d^2y}{dt^2} + x + y = 0.$$

Solution

The given equations can be written as

$$[D^2 - 3]x - 4y = 0; \quad x + [D^2 + 1]y = 0.$$

Eliminating y , we get

$$[(D^2 - 3)(D^2 + 1) + 4]x = 0, \text{ i.e., } (D^2 - 1)^2x = 0.$$

Auxiliary equation of the above differential equation in $\hat{x}$ is

$$(m^2 - 1)^2 = 0.$$

which gives $m = 1, -1, 1, -1$.

Hence

$$x = (A + Bt)e^t + (C + Dt)e^{-t}.$$

Putting this value of x in the first equation we get

$$y = \frac{1}{2}[B(1 - t) - A]e^t - \frac{1}{2}[C + (1 + t)D]e^{-t}.$$

Hence, $x = (A + Bt)e^t + (C + Dt)e^{-t}$

and $y = \frac{1}{2}[B(1 - t) - A]e^t - \frac{1}{2}[C + (1 + t)D]e^{-t}$ are the required solutions.

11. Solve the following system of differential equations:

$$\frac{dx}{dt} = 2y, \quad \frac{dy}{dt} = 2z, \quad \frac{dz}{dt} = 2x.$$

Solution

From the given equations, we get

$$\frac{d^3x}{dt^3} = 2 \frac{d^2y}{dt^2} = 4 \frac{dz}{dt} = 8x.$$

Solution of the equation $\frac{d^3x}{dt^3} - 8x = 0$ is $x = Ae^{2t} + e^{-t} [B \cos \sqrt{3}t + C \sin \sqrt{3}t]$.

So, $y = \frac{1}{2} \frac{dx}{dt} = Ae^{2t} + \frac{e^{-t}}{2} [-\sqrt{3}B \sin \sqrt{3}t + \sqrt{3}C \cos \sqrt{3}t - B \cos \sqrt{3}t - C \sin \sqrt{3}t]$

$$= Ae^{2t} + \frac{e^{-t}}{2} [(\sqrt{3}C - B) \cos \sqrt{3}t - (\sqrt{3}B + C) \sin \sqrt{3}t]$$

and $z = \frac{1}{2} \frac{dy}{dt} = Ae^{2t} + \frac{e^{-t}}{4} [-(\sqrt{3}C - B) \sqrt{3} \sin \sqrt{3}t - (\sqrt{3}B + C) \sqrt{3} \cos \sqrt{3}t - (\sqrt{3}C - B) \cos \sqrt{3}t + (\sqrt{3}B + C) \sin \sqrt{3}t]$

$$= Ae^{2t} + \frac{e^{-t}}{2} [(\sqrt{3}B - C) \sin \sqrt{3}t - (\sqrt{3}C + B) \cos \sqrt{3}t].$$

Hence $x = Ae^{2t} + e^{-t} [B \cos \sqrt{3}t + C \sin \sqrt{3}t],$

$$y = Ae^{2t} + \frac{e^{-t}}{2} [(\sqrt{3}C - B) \cos \sqrt{3}t - (\sqrt{3}B + C) \sin \sqrt{3}t]$$

and $z = Ae^{2t} + \frac{e^{-t}}{2} [(\sqrt{3}B - C) \sin \sqrt{3}t - (\sqrt{3}C + B) \cos \sqrt{3}t]$

are the required solutions.

12. Solve the following system of differential equations:

$$\frac{dx}{dt} = z + y, \quad \frac{dy}{dt} = x + z, \quad \frac{dz}{dt} = y + x.$$

Solution

Differentiating the first one and using all the equations in that we get

$$\frac{d^2x}{dt^2} - \frac{dx}{dt} - 2x = 0.$$

Solving this, we get

$$x = Ae^{-t} + Be^{2t}.$$

From the second and third equations, we get

$$\frac{d}{dt}(z - y) + z - y = 0.$$

Solving this, we get

$$z - y = Ce^{-t}.$$

Again we have

$$z + y = \frac{dx}{dt} = -Ae^{-t} + 2Be^{2t}.$$

Adding and subtracting we get

$$y = -\frac{A+C}{2} e^{-t} + Be^{2t}$$

and

$$z = \frac{C-A}{2} e^{-t} + Be^{2t}.$$

Hence, $x = Ae^{-t} + Be^{2t}, y = -\frac{A+C}{2} e^{-t} + Be^{2t}, z = \frac{C-A}{2} e^{-t} + Be^{2t}$ are the required solutions.

• Solve t

1. $\frac{dx}{dt} =$

3. $\frac{dx}{dt} =$

5. $\frac{dx}{dt} +$

7. $\frac{dx}{dt} +$

8. $\frac{dx}{dt} =$

10. $\frac{dx}{dt} =$

12. $\frac{dy}{dx} +$

14. $\frac{dx}{dt} =$

15. $4 \frac{dx}{dt}$

16. $\frac{dx}{dt}$

17. $\frac{dx}{dt}$

18. $\frac{dx}{dt}$

19. $3 \frac{a}{c}$

20. $\frac{d^2}{dt^2}$

21. $\frac{d^2}{dt^2}$

22. $\frac{d}{dt}$

23. $\frac{d}{dt}$